

Tutorial Notes:

Install ROS 2 + Gazebo

1. Preparation

```
sudo apt update
sudo apt install -y software-properties-common curl gnupg lsb-release
```

2. Add the ROS 2 apt repo

```
# Import the ROS key
sudo mkdir -p /usr/share/keyrings
curl -sSL https://raw.githubusercontent.com/ros/rosdistro/master/ros.key \
| sudo tee /usr/share/keyrings/ros-archive-keyring.gpg > /dev/null

# Add the repo
echo "deb [arch=$(dpkg --print-architecture)
signed-by=/usr/share/keyrings/ros-archive-keyring.gpg] \
http://packages.ros.org/ros2/ubuntu \
$(. /etc/os-release && echo $UBUNTU_CODENAME) main" \
| sudo tee /etc/apt/sources.list.d/ros2.list > /dev/null
```

3. Install ROS 2 Jazzy Desktop

```
sudo apt update
sudo apt install -y ros-jazzy-desktop
echo "source /opt/ros/jazzy/setup.bash" >> ~/.bashrc
```

4. Install Gazebo

```
# Make sure package lists are fresh
```

```
sudo apt update

# Install the full Gazebo + ROS 2 integration
sudo apt install -y ros-jazzy-ros-gz

# (Optional but handy GUI tool)
sudo apt install -y ros-jazzy-rqt-image-view
```

5. Settings

```
# source ROS 2 Jazzy
source /opt/ros/jazzy/setup.bash

# Use system Gazebo
export PATH=/usr/bin:$PATH

# Make gz aware of installed Gazebo commands (sim, gui, etc.)
export GZ_CONFIG_PATH=/usr/share/gz
```

Install and test PX4

Clone PX4-Autopilot

```
cd ~
sudo apt update
sudo apt install -y git

git clone --recursive https://github.com/PX4/PX4-Autopilot.git
```

Run the PX4 Ubuntu setup script

```
cd ~/PX4-Autopilot
bash ./Tools/setup/ubuntu.sh
source ~/.bashrc # if the script added env lines there
```

Test with the standard x500 model

```
cd ~/PX4-Autopilot
make px4_sitl gz_x500_mono_cam
```

Px4 settings

```
param set NAV_DLL_ACT 0
param set COM_RC_IN_MODE 1
param set COM_ARM_SDCARD 0
param set CBRK_SUPPLY_CHK 894281
param save
```

Arm and takeoff

```
commander arm
commander takeoff
```

Install and test Ground control

Install prerequisites:

```
sudo apt update
sudo apt install -y \
    libfuse2 \
    gstreamer1.0-plugins-bad gstreamer1.0-libav gstreamer1.0-gl \
    libxcb-xinerama0 libxkbcommon-x11-0 libxcb-cursor0
```

Download QGC:

Download [QGroundControl-x86_64.AppImage](#).

```
cd ~/Downloads
chmod +x QGroundControl-<arch>.AppImage
```

Open the QGC, it'll automatically connect with the px4

Test the camera

Test with the standard x500 model

Note: cd to px4 if you didn't already

```
cd ~/PX4-Autopilot
make px4_sitl gz_x500_mono_cam
```

Arm and takeoff

```
commander arm
commander takeoff
```

Let's build and launch a gazebo ros2 bridge

```
cat > ~/bridge_cam.yaml <<'YAML'
- gz_topic_name:
/world/default/model/x500_mono_cam_0/link/camera_link/sensor/camera/image
  ros_topic_name: /x500/camera/image_raw
  gz_type_name: gz.msgs.Image
  ros_type_name: sensor_msgs/msg/Image
  direction: GZ_TO_ROS

- gz_topic_name:
/world/default/model/x500_mono_cam_0/link/camera_link/sensor/camera/camera_
info
  ros_topic_name: /x500/camera/camera_info
  gz_type_name: gz.msgs.CameraInfo
  ros_type_name: sensor_msgs/msg/CameraInfo
  direction: GZ_TO_ROS
YAML
```

Next in the same terminal run the bridge:

```
ros2 run ros_gz_bridge parameter_bridge --ros-args -p  
config_file:=$HOME/bridge_cam.yaml
```

Let's see what the drone camera see:

```
ros2 run rqt_image_view rqt_image_view  
# Select: /x500/camera/image_raw
```